Development of a Functional Annotation and Taxonomic Classification Workflow for Metagenome-Assembled Genomes

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Abstract. While shotgun metagenomics enables the reconstruction of Metagenome-Assembled Genomes (MAGs), the functional annotation of communities comprising both prokaryotes and eukaryotes can be further optimized. To address this, an integrated Snakemake pipeline was applied for the comprehensive functional annotation of these MAGs. The architecture employs workflows adapted to each domain: Bakta, Prodigal, UPIMAPI, and reCOGnizer for prokaryotic MAGs, and MetaEuk and reCOGnizer with the KOG database for eukaryotic MAGs. All outputs converge into standardized, genome-specific Integrated Annotation Tables. Validating its utility, the workflow processed an anaerobic biomass dataset converting stearate to methane, generating detailed functional profiles that revealed prokaryotic metabolic and chemotactic networks. Concurrently, the eukaryotic module identified core cellular mechanisms, such as vesicle trafficking and signaling, allowing ecological inferences despite inherent assembly fragmentation. Ultimately, this approach produces multi-domain formatted outputs that seamlessly integrate into larger metagenomics frameworks, supporting downstream community metabolic modeling.

Keywords: Shotgun Metagenomics, Metagenome-Assembled Genomes, Multikingdom Microbiome, Bioinformatic Workflows, Functional Annotation

1 Contextualization and Motivation

Metagenomics transformed microbiome research by enabling the cultivation-independent analysis of microbial communities [26]. The transition from targeted amplicon sequencing (e.g., 16S rRNA) to shotgun metagenomics further advanced the field [14]. By sequencing all genetic material in a sample, shotgun metagenomics reveals both the taxonomic composition and functional potential across all domains of life. Crucially, it enables the computational reconstruction of MAGs, resolving strain-level identity and linking specific taxa directly to their metabolic capabilities [14].

In natural and engineered environments, prokaryotes and eukaryotes continuously interact to shape ecosystem function; consequently, metagenomic datasets naturally contain sequences from both domains [21,26]. Despite this multi-kingdom complexity, comprehensive workflows capable of analyzing the eukaryotic fraction remain scarce [28].

Although tools exist to analyze eukaryotic genomes, characterized by complex architectures like introns, alternative splicing, extensive non-coding regions, and repetitive elements, evaluating domains in isolation limits ecosystem understanding. Capturing true community dynamics and metabolic potential requires integrated pipelines that process both prokaryotic and eukaryotic MAGs [3,25]. However, applying standard, prokaryote-optimized annotation pipelines to whole-community datasets frequently results in the misclassification of eukaryotic sequences, the generation of false-positive gene predictions, or the complete exclusion of the eukaryotic fraction. While several analytical frameworks already exist to accommodate multi-kingdom MAG [6,13,25], managing this data complexity while minimizing information loss remains a continuous challenge. In this context, this project develops an annotation module to handle this complexity through independent pathways, providing standardized annotation outputs for downstream analyses such as community metabolic modeling [28].

2 Aims

The primary objective of this project is to develop and validate a modular bioinformatics workflow dedicated to the functional annotation of multi-kingdom genomes. This automated pipeline standardizes the processing of prokaryotic and eukaryotic organisms, addressing the current fragmentation in metagenomic analysis. This goal is supported by three specific objectives:

First, evaluating current computational tools for quality control and functional annotation. Second, developing a reproducible architecture that automatically routes sequences into tailored, domain-specific annotation modules. Finally, generating high-quality, standardized outputs that reveal microbiome functional potential and directly support downstream systems biology applications, specifically the reconstruction of Genome-scale Metabolic Models (GEMs).

3 State-of-the-art

3.1 The Concept of Metagenomics and MAGs

This section outlines the foundational concepts of modern microbiome analysis. It begins by defining the principles of metagenomics, subsequently exploring how advanced sequencing technologies enable the computational reconstruction of individual genomes directly from complex environmental samples.

Metagenomics is defined as the culture-independent analysis of genetic material recovered directly from environmental or host-associated samples. By bypassing the need for isolation and cultivation in the laboratory, metagenomic approaches enable the study of complex microbial communities in their native contexts. This capability enables the simultaneous capture of the taxonomic composition and functional potential of the community as a whole [26]. Consequently, metagenomics has established itself as a central methodology in microbiome research, with wide-ranging applications in environmental sciences, biotechnology, agriculture, and human health.

The true power of this approach has been cemented by the adoption of Next-Generation Sequencing (NGS) technologies, which form the basis of shotgun metagenomics [4,9,19]. By sequencing the entirety of Deoxyribonucleic acid (DNA) extracted from a sample, shotgun metagenomics enables a comprehensive analysis that encompasses all domains of life (bacteria, archaea, and eukaryotes). More importantly, this technique enables the computational reconstruction of complete or partial genomes from a mixture of sequences. These reconstructed genomes, known as MAGs, have become the fundamental unit in metagenomics, enabling the direct detection of genes, operons, and metabolic pathways of individual populations within the ecosystem [14].

3.2 Overview of Multi-Kingdom Genomic Analysis

This section outlines the computational challenges inherent to analyzing samples containing multiple biological domains. It contrasts the genomic complexities of prokaryotes and eukaryotes, evaluates the limitations of current integration strategies, and establishes the need for modular routing architectures.

Shotgun metagenomics has the major advantage of capturing all DNA in a sample, including bacteria, archaea, and eukaryotes [14]. However, simultaneously processing prokaryotic and eukaryotic sequences poses a major computational challenge. The central problem lies in the fact that most current bioinformatics tools have been designed and optimized solely for prokaryotic organisms, whose genomes are typically more compact and simpler.

In contrast, the genomes of eukaryotes, such as fungi and protists, are much more complex and challenging. They are substantially larger and contain specific structures, such as introns, vast non-coding regions, and alternative splicing mechanisms [8]. If we apply the algorithms developed for bacteria directly to these eukaryotic genomes, the results will be incorrect. Applying prokaryotic gene prediction algorithms to eukaryotic sequences often leads to structural mispredictions, as these tools fail to recognize introns. Consequently, a single eukaryotic gene is erroneously split into multiple independent coding sequences. This artificial fragmentation leads to an inflated gene count and significantly compromises the accuracy of downstream metabolic pathway reconstructions.

While specialized software has been successfully developed to annotate these complex eukaryotic genomes independently, the primary obstacle in modern metagenomics remains the integration of these distinct biological domains. Although specific tools have been developed to perform this integration, they present significant compromises. Current approaches typically fall into two categories. The first employs generalized unified workflows, such as SqueezeMeta [22]. Although these pipelines offer the advantage of streamlined execution by processing all data simultaneously, they frequently suffer from a strong prokaryotic bias, relying on generalized algorithms that fail to capture complex eukaryotic structures. The second category involves modular sequential processing, using platforms like Anvi'o [3] or MetaWRAP [23]. Although integrating specialized tools can yield high accuracy, it often requires complex manual interventions and presents challenges in standardizing diverse output formats. Furthermore, traditional pipelines frequently lack true modularity, restricting the ability of both developers and end-users to easily swap, update, or remove specific analytical components. Consequently, investigating cross-kingdom ecological interactions remains a complex methodological challenge.

To address these integration challenges, it is becoming increasingly clear that all DNA within a sample cannot be treated uniformly. Overcoming current obstacles requires a paradigm shift towards modular architectures capable of processing genomes that have been previously classified by biological domains. Developing workflows that can act as dedicated routing engines, seamlessly directing pre-segregated prokaryotic and eukaryotic sequences to their respective specialized annotation tools, represents a critical gap in the current literature. Filling this gap is essential to maximize the biological accuracy of comprehensive multi-kingdom microbiome studies.

3.3 Taxonomic and Functional Annotation Strategies

This section details the distinct analytical strategies required for the taxonomic classification and functional annotation of separated genomic data. It first examines the highly optimized workflows utilized for prokaryotic organisms, subsequently exploring the sophisticated computational tools necessary to resolve complex eukaryotic structures. Finally, the section evaluates current comprehensive pipelines attempting to integrate these domains, highlighting their respective advantages and limitations.

Following the successful segregation of MAGs by biological domain [25], the next step involves a thorough characterization of their genetic content. In the context of metagenomics, this process unfolds along two fundamental axes: taxonomic classification, which identifies evolutionary lineages, and functional annotation, which elucidates metabolic potential. Given the marked divergence in genomic architecture between prokaryotes and eukaryotes, it is imperative to adopt independent and specialized analytical strategies for each group.

Prokaryotic Genomes Regarding the prokaryotic domains of bacteria and archaea, bioinformatic workflows are currently highly optimized. While taxonomic classification is frequently anchored by GTDB-Tk [2] as the established methodological standard, alternative frameworks, including the Bin Annotation Tool [24] and PhyloPhlAn [1], provide robust complementary strategies for phylogenetic profiling. Instead of relying on a single marker gene such as 16S rRNA, these modern approaches use the alignment of dozens of universal marker proteins, enabling the precise placement of the MAGs within a robust phylogenetic tree topologically consistent with the Genome Taxonomy Database (GTDB).

Concerning functional annotation, although Prokka [17] has historically stood out for its efficiency, contemporary pipelines such as Bakta [16] are increasingly adopted for their enhanced feature resolution. By taking advantage of the compact genomic architecture of bacteria, specifically the general absence of introns, these tools integrate fast-running *ab initio* gene prediction algorithms, most notably Prodigal [5], with extensively curated databases. To achieve a more exhaustive functional characterization, these baseline annotations are frequently complemented with specialized downstream tools. For instance, UPIMAPI [18] facilitates the comprehensive retrieval of functional information by mapping sequence identifiers to the UniProt database, whereas reCOGnizer [18] enables targeted, domain-based profiling through the Clusters of Orthologous Groups (COG) database. Together, these frameworks generate in-depth functional profiles that directly support advanced downstream systems biology applications.

Eukaryotic Genomes In contrast, the annotation of eukaryotic MAGs, namely eukaryotes, requires a substantially more sophisticated computational approach. Traditional predictive tools often struggle with the discontinuity of eukaryotic genes fragmented by introns. To overcome this limitation, modern workflows rely on specialized solutions.For structural and functional annotation, MetaEuk [8] offers a dynamic sequence similarity-based strategy to accurately identify coding exons and infer intron boundaries directly from assembled contigs. Alternatively, comprehensive workflows like EukMetaSanity [12] utilize *ab initio* gene predictors, including Augustus [20], to resolve these complex genomic architectures. Once putative protein-coding sequences are identified, achieving a deeper functional resolution requires targeted profiling. In this context, tools such as reCOGnizer [18] can be leveraged using the euKaryotic Orthologous Groups (KOG) database, enabling accurate domain-based functional categorization of eukaryotic proteins. Finally, beyond gene prediction and annotation, ensuring the overall validity of eukaryotic MAGs demands rigorous quality control and lineage assessment. Estimating genome completeness and contamination is a crucial validation step effectively managed by dedicated tools such as EukCC [15] and BUSCO [10].

Ultimately, while the independent annotation of eukaryotic and prokaryotic fractions is well established, the primary obstacle lies in their seamless integration. Currently, researchers rely on a few comprehensive pipelines that attempt to process communities with multiple domains, but these solutions present distinct compromises between computational efficiency and biological accuracy. For instance, unified pipelines such as SqueezeMeta [22] offer rapid joint assembly profiling, but they typically rely on generalized gene prediction strategies (e.g., Prodigal) that may not fully capture the complex structural features of eukaryotic genomes [8]. Other approaches utilize modular sequential processing, such as MetaWRAP [23], which provides an excellent framework for prokaryotic binning and refinement, yet its native support for eukaryotes remains limited due to its reliance on prokaryote-specific tools like CheckM and Prokka [15]. In contrast, highly modular platforms like Anvi'o [3] provide robust and dedicated workflows for both domains, although comparative reviews indicate that their interactive nature often demands significant computational resources and extensive manual orchestration [27].

Table 1. Critical comparison of current bioinformatics pipelines for the integration of multiple biological domains.

Pipeline	Integration Strategy	Key Advantages	Major Limitations
SqueezeMeta [22]	Unified joint assembly and homology search.	Highly automated; fast execution; simultaneously profiles prokaryotes and eukaryotes.	Shallow structural annotation for eukaryotes (e.g., relies on prokaryotic gene callers); relies heavily on reference databases **[8]**.
MetaWRAP [23]	Modular sequential processing.	Excellent for prokaryotic binning and refinement; highly flexible framework.	Eukaryotic support is limited (reliance on CheckM and Prokka) and often requires unofficial custom adaptations **[15]**.
Anvi'o [3]	Highly modular and interactive architecture.	Deep specialized analysis for both domains; excellent visualization capabilities.	Steep learning curve; demands intensive computational resources and manual orchestration **[27]**.
nf-core/mag [7]	Community-curated hybrid assembly and binning (Nextflow).	Highly reproducible and standardized; explicitly integrates eukaryotic tools (Tiara, MetaEuk, BUSCO).	Generalist framework focused primarily on upstream MAG recovery rather than deep functional routing for metabolic modeling **[7]**.
metaGEM [28]	Snakemake workflow linking MAG recovery to metabolic modeling.	Predicts metabolic interactions directly from metagenomes; includes supplementary modules for eukaryotic binning.	Core architecture is heavily optimized for bacterial communities, treating eukaryotic recovery as a secondary step rather than fully integrated **[28]**.

4 Methodology

This work contributes to a broader pipeline for MAGs analysis developed by the BiSBi research group at the University of Minho. Within this broader framework, the present report focuses on the design and integration of the gene prediction and functional annotation module.

To address current multi-kingdom integration challenges and meet established objectives, this work proposes a modular, reproducible methodology that integrates an optimized gene prediction and functional annotation module into the global pipeline. Figure 2 illustrates the architecture and data flow, which begins by processing FASTA-formatted MAGs.

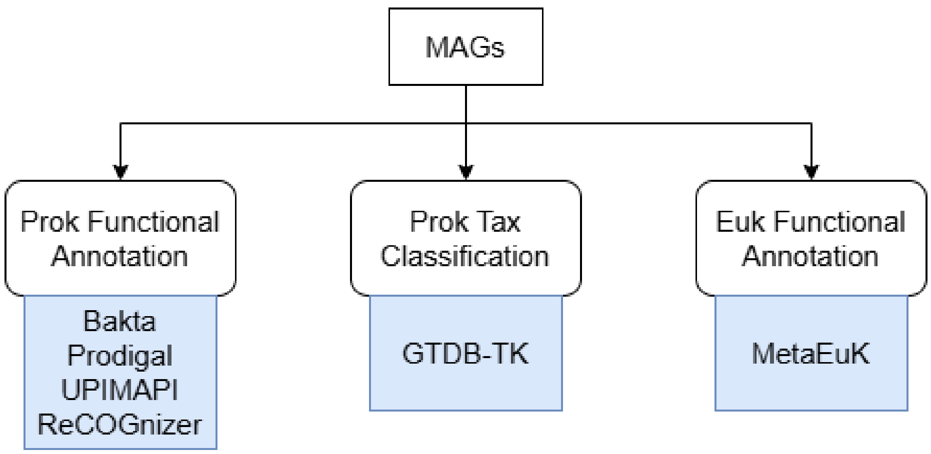

Fig. 1. Architectural overview of the metagenomic pipeline. Metagenome-Assembled Genomes (MAGs) are routed according to their domain: prokaryotic bins undergo taxonomic and functional annotation (GTDB-Tk, Prodigal, Bakta, UPIMAPI, and reCOGnizer), while eukaryotic bins are processed via MetaEuk, with all data converging into a centralized Integrated Annotation Table.

4.1 Workflow Orchestration and Execution

To orchestrate parallel processing and integrate seamlessly into the group's infrastructure, the pipeline uses Snakemake [11] for workflow management. This Python-based tool models file dependencies through a directed acyclic graph, enabling automatic task parallelization. To guarantee full reproducibility and prevent software conflicts across diverse infrastructures, Snakemake integrates natively with Conda and Docker for strict environment isolation. Relying on these community-tested components minimizes custom scripting, reduces human error, and ensures standardized, reliable execution.

4.2 Software Integration and Wrapper Development

The majority of command-line bioinformatics tools do not communicate natively with the data management and execution systems required by the Snakemake architecture. To ensure a seamless, modular, and highly extensible integration of the annotation process into the global pipeline, it was necessary to develop custom wrappers for a wide range of fundamental programs. Specifically, integration modules were designed and implemented for the tools MetaEuk, Bakta, and Prodigal (gene prediction), reCOGnizer and UPIMAPI (functional annotation). Additionally, wrappers for Prokka and GTDB-Tk were developed to provide methodological alternatives and ensure backward compatibility. However, these tools were not integrated into the primary workflow of this study. Prokka was superseded by Bakta due to its more comprehensively updated databases and enhanced annotation capabilities. Furthermore, GTDB-Tk was excluded from the core pipeline because of its demanding computational memory requirements, which would limit the overall accessibility and scalability of the framework.

The design and structuring of this integration ecosystem were developed in strict accordance with the guidelines and standards established by the official Snakemake Wrapper Repository. This open-access catalog played a central role in the methodological development, serving as the primary architectural reference for the creation of custom modules. The comprehensive study of the official Snakemake Wrappers documentation, alongside the rigorous consultation and analysis of dozens of standardized wrappers available in the repository, enabled the assimilation of best practices in software encapsulation. This approach ensured that the custom-built code met the standards of quality, robustness, error handling, and log management demanded by the scientific community, thereby facilitating the future scalability and maintenance of the project.

Based on this reference and the official documentation of each tool, the encapsulation of each program required the coordinated creation of three essential components. First, the workflow rules (in .smk format) were established, which consist of the rigorous definition of input files, expected outputs, tool-specific parameters, and the allocation of computational resources. Next, to ensure the complete reproducibility of the study, configuration files (in .yaml format) were developed, instructing the Conda manager to create isolated virtual environments containing the exact versions of each program and their respective dependencies. Simultaneously, execution scripts were written in Python (.py), whose central function is to interpret the objects generated by Snakemake and dynamically translate them into properly formatted terminal commands.

4.3 Workflow Orchestration and Execution Dynamics

With the individual wrappers properly developed and tested, the next step consisted of orchestrating these tools into a continuous and coherent workflow. The structuring of the main workflow was not designed arbitrarily, but rather based on the Snakemake Workflow Template, a standardized repository widely validated by the community. The adoption of this architectural model ensures that the project follows the most rigorous guidelines for directory organization—clearly separating configuration logic, execution rules, scripts, and virtual environments. In addition to promoting best development practices, this standardization proved to be a crucial strategic step in ensuring the modularity and interoperability of the code, facilitating the seamless integration of this annotation module into the global processing infrastructure developed by the research group.

The execution dynamics of the pipeline and its external communication are governed by two central files: the global configuration file (config.yaml), which dictates essential execution parameters such as database paths, thread allocation, and output directories, alongside the sample mapping file,(sample.tsv). The sample.tsv file assumes particular importance, as it acts as the direct bridge between the annotation module and the upstream phases of the research group's pipeline. Generated during the upstream processing phase, this tabular document systematically organizes the input data through three essential columns. These specify the unique identifier for each MAG, the directory path to the corresponding FASTA sequence, and the biological domain previously assigned by the Tiara classification tool. This domain assignment is crucial for automatically routing the genomes into their respective annotation modules. This decoupled architecture ensures that the annotation module can be continuously fed with new data without the need to rewrite any part of the source code.

It is based on the combined assessment of the sample mapping file and a complementary quality report table that Snakemake orchestrates the execution. This quality table contains metrics previously calculated by CheckM2 for prokaryotes and EukCC for eukaryotes, allowing the pipeline to systematically filter out low-quality assemblies. Ultimately, this rigorous evaluation enables the accurate sorting and branching of the remaining high-quality MAGs prior to any annotation, automatically directing them into distinct analytical pathways.

Prokaryotic Pathway : As illustrated in Figure 2, if the sample belongs to the prokaryotic domain, the workflow processes the data through a sequential two-tiered strategy. Initially, the original FASTA sequences are routed to Bakta [16] and Prodigal [5] for high-resolution structural annotation and gene prediction. The predicted protein sequences generated in this first step are subsequently used as the direct input for UPIMAPI [18] and reCOGnizer [18], expanding the functional baseline through comprehensive profiling against UniProt and COG databases.

Eukaryotic Pathway : On the other hand, if the sample is eukaryotic, the pipeline triggers modules designed for the complexity of these genomes. It utilizes MetaEuk [8] to accurately predict complex genetic structures, overcoming the discontinuity of introns. Subsequently, reCOGnizer utilizes the KOG database for precise functional categorization.

Following the parallel processing in these two branches, the final stage of the pipeline concatenates the domain-specific annotations into a unified dataset, ensuring direct compatibility with downstream community metabolic modeling tools.The end result is the generation of standardized functional annotations in unified data tables. These consolidated outputs are structured for direct integration into downstream computational applications, specifically enabling the reconstruction and simulation of GEMs for entire biological communities.

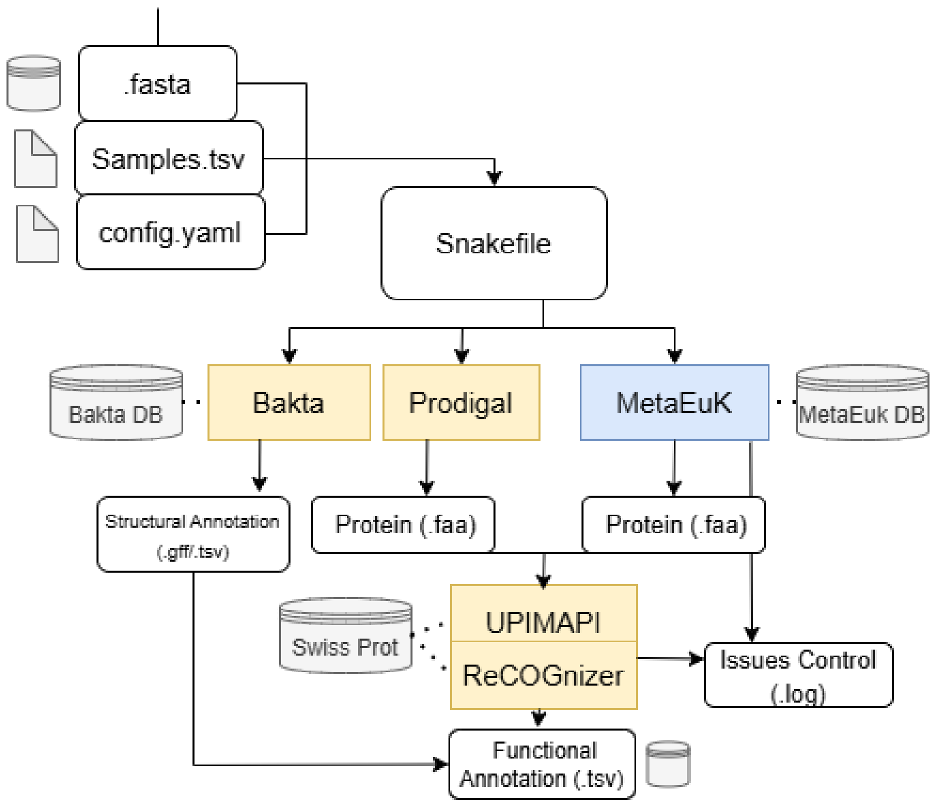

Fig. 2. Architectural overview of the annotation workflow. Input files are processed via a central Snakefile, routing data through domain-specific tools: Bakta for structural annotation, and Prodigal and MetaEuk for protein prediction. The resulting protein sequences are functionally mapped using UPIMAPI and reCOGnizer, converging with the structural data into a final Functional Annotation file.

4.4 Pipeline Validation and Dataset Description

To validate the biological utility of the developed module, the computational framework was tested using a real environmental metagenomic dataset. This sample was derived from suspended anaerobic biomass, originally collected from an anaerobic digester and subsequently incubated under anaerobic conditions, using stearate as the sole carbon and energy source for methanogenic conversion. The validation test was performed by executing the Snakemake-based workflow on the pre-assembled MAGs from this community, seamlessly routing the data through the respective domain-specific annotation modules to generate the final integrated profiles.

5 Results and Discussion

This section presents the main results derived from the computational annotation pipeline, focusing on both the structural characteristics of the genome and the functional potential of the recovered microbial population.

5.1 Structural Annotation of Prokaryotes and Eukaryotes

The workflow demonstrated versatility by autonomously processing MAGs from different domains. The structural annotation of the prokaryotic MAGs was conducted in a standardized manner using Bakta, allowing for the consistent extraction of genomic metrics such as size, GC content, N50, and the identification of coding sequences (CDSs) and non-coding elements. In parallel, the annotation of eukaryotic samples was directed to MetaEuk.

The structural processing step successfully reconstructed the dataset, with the detailed metrics for representative Metagenome-Assembled Genomes presented in Table 2. Due to the biological nature of the micro-eukaryotic population, the initial MAGs recovered from this fraction presented suboptimal assembly quality, yielding fewer than 500 predicted coding sequences per bin. Consequently, to rigorously validate the eukaryotic processing branch of the pipeline despite environmental fragmentation, an external high-quality fungal isolate, *Paecilomyces variotii*, was introduced as a positive control. The submission of this control genome to MetaEuk resulted in the successful identification of 8230 predicted coding sequences, a value consistent with the reference genomic baseline for this species. Furthermore, downstream functional annotation successfully captured its core eukaryotic repertoire, including essential housekeeping genes, structural proteins, and central metabolic pathways, empirically demonstrating the workflow's capacity to not only recover but accurately annotate key functional elements.

5.2 Functional Integration and Master Table Creation

To overcome the challenge of biological data dispersion common in annotation pipelines, the structural and functional results generated by the different algorithms were consolidated into standalone, genome-specific tabular resources.

Table 2. Structural assembly and annotation metrics for the analyzed Metagenome-Assembled Genomes.

MAG Identifier	Contigs	Size (Mb)	GC (%)	Identified CDSs
`EST6_prok_bin_001`	236	5.18	57.99	4686
`EST6_prok_bin_040`	281	3.32	63.83	3330
`EST6_euk_bin_002`	246	0.32	55.44	427

Rather than creating a single global document, the pipeline generates an individual Functional Master Table for each processed MAG.

Using the representative genome `EST6_prok_bin_040` as a case study, the genomic foundation of its specific Master Table relied on the 3330 predicted protein sequences initially generated by Prodigal. These amino acid sequences served as the standardized input for both structural and functional downstream tools. Structurally, Bakta processed these sequences to extract the fundamental features of the prokaryotic genome. Simultaneously, two layers of functional annotation were integrated using the same Prodigal-derived baseline. The sequences were annotated against the UniProt database via UPIMAPI, obtaining direct homology mapping, while the reCOGnizer algorithm mapped functions and KEGG Orthology identifiers within the Clusters of Orthologous Groups framework.

The results from these three tools were subsequently merged, generating the final genome-specific Master Table. This table integrates the baseline structural information provided by Bakta with the functional data for each gene, specifically encompassing the UniProt homology information, such as protein names and identifiers, alongside the robust orthologous classification from reCOGnizer. This modular data centralization proved to be fundamental, not only by increasing the reliability of the annotation through the cross-referencing of different databases but also by enabling the rapid filtering of metabolic pathways and functions of ecological interest for any specific member of the microbial community.

5.3 Biological Exploration: Functional Profiling of a Target Metagenome-Assembled Genome

The profound utility of the consolidated functional data can be explicitly illustrated through the in-depth analysis of `EST6_prok_bin_040`. Crucially, the identification of multiple genes encoding *Acyl-CoA dehydrogenase* perfectly corroborates the ecological niche of this organism; this enzyme is a key catalyst in the β-oxidation of fatty acids, directly aligning with the stearate-fed conditions of the anaerobic digester. Furthermore, the high abundance of *Histidine kinases* and *Sigma-54* transcription regulators reveals robust environmental signaling and response capabilities.

Direct data filtering also allowed for the identification of an extensive block of genes dedicated to cellular mobility. The annotation confirmed the presence of the structural and regulatory machinery for bacterial flagellum biogenesis

Table 3. Most frequent annotated protein functions and their associated biological pathways identified within the representative genome `EST6_prok_bin_040`.

Annotated Protein/Enzyme	**Gene Count**	**Associated Biological Pathway**
Histidine kinase	31	Signal Transduction (Two-component system)
Lipoprotein	16	Cell Wall and Membrane Biogenesis
Tetratricopeptide repeat protein	10	Protein-protein Interactions / Scaffolding
Sigma-54-dependent regulator	9	Transcription Regulation / Stress Response
Sialic acid TRAP transporter SiaT	9	Substrate Membrane Transport
Acyl-CoA dehydrogenase	8	Fatty Acid Degradation (β-oxidation)
ABC transporter ATP-binding protein	8	Active Transmembrane Transport
4Fe-4S ferredoxin-type domain	8	Electron Transport / Energy Metabolism
Prepilin-type cleavage domain	7	Cellular Mobility (Type IV pili assembly)

(*flg*, *fli*, and *flh* gene families), as well as the motor complexes responsible for rotation (*motA* and *motB*). Together with the assembly of Type IV pili, this suggests that this specific bacterial population possesses an active capacity for chemotaxis within its microenvironment.

Concurrently, eukaryotic functional annotation further detailed community complexity. Despite assembly limitations, identifying genes for vesicle trafficking, protein glycosylation, and cytoskeleton organization highlighted essential structural maintenance mechanisms. Ultimately, this integrated multi-domain approach facilitates deeper ecological inference, capturing both active bacterial lifestyles and specialized eukaryotic cellular processes within this anaerobic community.

6 Conclusion

The present study served primarily as a proof of concept for the integration of multiple genomic annotation tools into a structured and reproducible computational pipeline. We successfully applied Bakta, Prodigal, MetaEuk, UPIMAPI, and reCOGnizer to environmental samples from distinct domains (bacteria and eukaryotes), demonstrating the feasibility of orchestrating these algorithms to extract meaningful biological and ecological patterns. The consolidation of data into a Functional Master Table proved to be a highly effective strategy, as illustrated by the MAG 040 case study, enabling a clear interpretation of the population's cellular mobility and signaling dynamics.

However, our analysis highlighted the intrinsic limitations of current metagenomic approaches. The efficacy of functional characterization is constrained by the fragmentation inherent in MAG assemblies and the reliance on reference databases, which is reflected in the considerable fraction of hypothetical proteins that remain uncharacterized. Furthermore, focusing on a specific environmental sample served the purpose of methodological validation but restricts ecological extrapolation on a global scale.

With this in mind, future applications of this workflow should aim to expand processing to larger-scale and more diverse metagenomic communities. It will also be crucial to integrate automated visualization modules and explore the mapping of complete metabolic pathways, thereby allowing for a more precise understanding of the ecological roles and interactions of uncultivated species within their natural habitats.

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